

Robust Fake Review Detection with Large Language Models: A Prompt Rewording and Review Length Analysis

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Abstract

Automated detection of deceptive online reviews is essential for maintaining trust in digital marketplaces, yet the robustness of large language model (LLM) classifiers for this task remains poorly understood. Although modern NLP systems perform well on benchmark datasets, their predictions may be sensitive to prompt formulation and input characteristics, raising concerns about reliability in real-world moderation systems. This study investigates the robustness of LLM-based fake review detection by analyzing prediction stability under prompt rewording and performance variation across review lengths. Using the *difraud/difraud* dataset of labeled business-domain reviews, we evaluate semantically equivalent prompt templates created through controlled rewording of a base classification prompt and analyze performance across short, medium, and long reviews using token-count quantile thresholds. Experiments compare baseline machine learning and transformer-based classifiers using accuracy, precision, recall, and F1-score, with emphasis on recall and F1-score due to the high cost of undetected deceptive reviews. The results provide insights into the stability and limitations of LLM-based deception detection systems.

1 Introduction

Online consumer reviews strongly influence purchasing decisions, but deceptive reviews present a major challenge for digital marketplaces. Detecting such content is a natural language processing (NLP) task that analyzes review text to determine whether it reflects genuine or misleading experiences (Miah et al., 2025). As review volumes exceed the capacity of human moderation, automated fake review detection has become an important research problem. Recent advances in large language models (LLMs) have improved performance across NLP tasks such as text classification (Jañez-Martino et al., 2024), sentiment analysis (Miah et al., 2025), and misinformation detection (Trinh et al., 2025). Transformer architectures and

prompt-based learning enable models to capture complex linguistic patterns in text (Schick and Schütze, 2021; Wei et al., 2022). However, the robustness of LLM-based classifiers remains uncertain, as predictions may vary under prompt rewording or small input changes (Zhao et al., 2021). Such instability is particularly problematic for deceptive review detection, where reliable predictions are essential for automated moderation (Miah et al., 2025). Prompt rewording or differences in review length may therefore expose weaknesses in model robustness and generalization (Wang et al., 2022). In this work, we evaluate the robustness of LLM-based fake review detection by analyzing prediction stability under prompt rewording and performance across review lengths using quantile-based token thresholds. We compare supervised transformer classifiers and prompt-based LLM models on the *difraud/difraud* dataset using standard metrics, with emphasis on recall and F1-score.

2 Related Work

Prior work on deceptive language detection examines how linguistic patterns reveal misleading online content (Ott et al., 2011). Recent studies show that large language models (LLMs) can detect deception across domains and modalities, including multimodal settings (Miah et al., 2025), cross-cultural deceptive text (Papantoniou et al., 2025), and stylistic differences between human- and machine-generated deception (Trinh et al., 2025). However, most studies emphasize classification accuracy rather than robustness to input variations. Robustness evaluation has therefore become an important area of NLP research. (Wang et al., 2022) survey robustness evaluation methods, while prior work shows that LLM predictions can vary substantially depending on prompt formulation (Zhao et al., 2021). Prompt-based learning enables pretrained models to perform classification through natural-language prompts (Schick and Schütze, 2021), but such methods remain sensitive to prompt wording (Zhao et al., 2021). Motivated

by these findings, our work evaluates prediction stability under prompt rewording and analyzes the effect of review length on fake review detection.

2.1 Methodologically Similar ACL Papers

Several ACL papers inform our methodological design. (Devlin et al., 2019) establish a standard framework for fine-tuning transformer models for text classification, motivating our supervised baseline. (Schick and Schütze, 2021) demonstrate prompt-based classification using pattern-exploiting training (PET), which informs our prompt-based evaluation setup. Additionally, (Zhao et al., 2021) show that language model predictions vary under prompt formulation, motivating our robustness analysis of prompt rewording.

3 Methodology

3.1 Dataset and Task Definition

We use the *DiFraud* (Domain Independent Fraud Detection) benchmark dataset, which contains labeled customer reviews for deception detection in the business domain. The dataset is publicly available through the Hugging Face repository and consists primarily of Amazon product reviews. Each instance contains review text and a binary label indicating whether the review is real (0) or fake (1). The dataset includes 16,776 training samples, 2,097 validation samples, and 2,098 test samples, resulting in over 20,000 labeled reviews with a nearly balanced class distribution. Fake review detection is formulated as a document-level binary classification task where the input x is the review text and the output y indicates whether the review is deceptive or genuine. The dataset has been cleaned using the Cleanlab framework to remove duplicates, extreme-length entries, and noisy samples. Although well curated, the dataset may contain stylistic biases because reviews originate from benchmark collections rather than live platforms, and the proportion of fake reviews may differ from real-world distributions.

3.2 Data Preprocessing

The dataset was loaded using the Hugging Face *Datasets* library and processed in Python. To support the review-length analysis, token counts were computed for each review using the RoBERTa tokenizer. Based on the token-count distribution in the training set, quantile thresholds were used to divide the dataset into three review-length groups. The

33rd percentile corresponds to 42 tokens and the 66th percentile corresponds to 71 tokens. Reviews were categorized as short (≤ 42 tokens), medium (43–71 tokens), or long (> 71 tokens). Each review was assigned to one of these groups to enable performance comparisons across different input lengths.

3.3 Model Selection

Two model types are evaluated: a supervised transformer classifier and a prompt-based large language model (LLM). The supervised baseline is *RoBERTa-base*, which is fine-tuned on the *DiFraud* training set to learn linguistic patterns associated with deceptive reviews. For the LLM evaluation, we use the instruction-tuned *Qwen2.5-7B-Instruct* model in a zero-shot setting, where reviews are embedded in prompts that instruct the model to label them as real or fake. Preliminary experiments with *Qwen2.5-3B-Instruct* produced less consistent predictions, so the 7B model was selected. Zero-shot prompting was chosen since fake review detection is a direct classification task. These models enable comparison between supervised transformer and prompt-based LLM approaches.

3.4 System Pipeline

The experimental pipeline converts raw review text into predicted labels through several processing stages. Review text and labels are first loaded from the *DiFraud* dataset. Each review is tokenized using the RoBERTa tokenizer, and token counts are used to assign reviews to short, medium, or long length groups based on quantile thresholds computed from the training set.

For the supervised baseline, the tokenized dataset is used to fine-tune a *RoBERTa-base* model for binary classification. In parallel, the *Qwen2.5-7B-Instruct* model performs zero-shot classification, where each review is embedded within a prompt instructing the model to label it as real or fake. The prompt used is: “You are an expert at detecting deceptive online reviews. Determine whether the following review is Real (written by a genuine customer) or Fake (deceptive or promotional). Respond with only one word: Real or Fake.”

The outputs of both models are parsed to extract predicted labels, which are then compared with the ground-truth labels and evaluated using the metrics described in Section 5. The overall workflow of the system is illustrated in Figure 1.

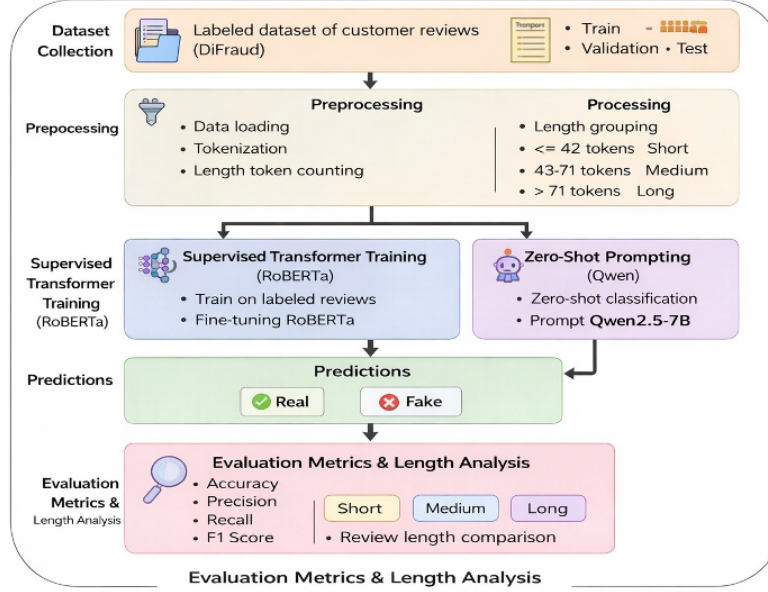


Figure 1: Overview of the system pipeline.

3.5 Compute and Feasibility

All experiments were implemented in Python using the Hugging Face Transformers and Datasets libraries and executed in Google Colab with GPU acceleration. The *RoBERTa-base* model was fine-tuned for two epochs using a batch size of 16 and a learning rate of 2×10^{-5} .

For the LLM experiments, the *Qwen2.5-7B-Instruct* model was executed locally using 4-bit quantization to reduce memory usage and enable inference on GPUs with limited memory capacity. Reviews were processed sequentially to avoid GPU memory overflow while maintaining reproducible evaluation across the full test dataset.

4 Evaluation Framework

To evaluate the transformer models, large language models (LLMs), and prompting techniques used in our fake review detection project, we employ standard quantitative evaluation metrics together with a review-length analysis. Fake review detection is formulated as a binary classification task and evaluated using **accuracy**, **precision**, **recall**, and **F1-score**.

Accuracy measures overall prediction correctness, while precision and recall capture different aspects of detection performance. Recall, defined as the proportion of fake reviews correctly identified among all deceptive reviews, is particularly important because false negatives (fake reviews

predicted as genuine) are the most harmful outcome in moderation systems. The F1-score, the harmonic mean of precision and recall, summarizes the trade-off between detecting deceptive reviews and avoiding false positives. We also perform a review-length analysis by grouping reviews into short, medium, and long categories using token-count quantile thresholds and evaluating performance within each group to assess how review length influences classification results, particularly recall.

4.1 Baselines

The evaluation also incorporates baseline comparison and robustness analysis. We use *RoBERTa* as our baseline model, motivated by prior work comparing transformer-based models for spam email detection (Jañez-Martino et al., 2024). The performance of prompt-based large language models (LLMs), including zero-shot and few-shot prompting configurations, is compared against this supervised baseline to determine whether prompting-based approaches can achieve comparable or improved detection performance.

4.2 Evaluation Strategy

The dataset is divided into training, validation, and test splits using an 80%/10%/10% partition. The dataset contains 20,971 Amazon reviews, including 10,492 deceptive and 10,479 non-deceptive samples, resulting in a nearly balanced class distribu-

tion. The baseline *RoBERTa* model is trained on the training split, and performance is evaluated on the test set using recall, precision, accuracy, and F1-score across each review-length group. These results serve as a reference for comparison with large language model classifiers, specifically the *Qwen* models.

5 Experiments and Progress

5.1 Experimental Setup

All experiments were implemented using the Hugging Face Transformers library and executed in Google Colab with GPU acceleration. Model performance was evaluated using standard classification metrics (accuracy, precision, recall, and F1-score) computed with the *scikit-learn* library, with recall treated as the primary metric due to the importance of detecting deceptive reviews.

5.2 Experiment 1: Baseline Analysis

The first experiment establishes a supervised baseline model using the *RoBERTa*-base architecture. The model was fine-tuned for a binary classification task, where each review is labeled as either fake (1) or real (0). Prior to training, the review texts were tokenized using the *RoBERTa* tokenizer with truncation and padding applied to a maximum sequence length of 512 tokens, ensuring that all inputs have a consistent format compatible with the transformer architecture.

The model was trained using the Hugging Face Trainer API and evaluated on the test dataset after training. The hyperparameters used during fine-tuning are shown in Table 1.

Parameter	Value
Model	RoBERTa-base
Maximum sequence length	512
Learning rate	2×10^{-5}
Training batch size	16
Evaluation batch size	16
Number of epochs	2
Weight decay	0.01

Table 1: Hyperparameters used for fine-tuning the *RoBERTa* baseline model.

During training, the model parameters were optimized on the training dataset while performance was monitored on the validation set. After training, the final model was evaluated on the test dataset to assess its ability to distinguish between fake and real reviews. In addition to overall performance,

the evaluation was conducted separately across the three review length groups (short, medium, and long) in order to analyze how the amount of textual information influences classification performance.

Length Group	N	Accuracy	Precision	Recall	F1
Short	699	0.70	0.66	0.69	0.68
Medium	677	0.70	0.72	0.84	0.77
Long	722	0.75	0.73	0.71	0.72

Table 2: *RoBERTa* performance across review length groups.

The results shown in Table 2 indicate that the *RoBERTa* model performs consistently across different review lengths. The highest recall is observed for medium-length reviews, suggesting that reviews with a moderate amount of detail provide clearer signals for detecting deceptive patterns. Short reviews contain less contextual information, which may make it more difficult for the model to identify deceptive cues, while long reviews may introduce additional noise that slightly affects classification performance.

5.3 Experiment 2: Improved Method or Comparative Analysis

The second experiment evaluates large language model (LLM) performance for fake review detection using zero-shot prompting. The *Qwen2.5-7B-Instruct* model receives each review embedded in a classification prompt instructing the model to label the review as genuine or deceptive. Inputs are limited to 1024 tokens, and the generated output is parsed to obtain a binary prediction. Because the model is used in a zero-shot setting, predictions rely entirely on the model’s pre-trained knowledge. To ensure comparability with the supervised baseline, the same preprocessing steps and review-length groupings were used. Predictions on the test dataset were evaluated using the same metrics as the *RoBERTa* baseline, and performance was analyzed across short, medium, and long review groups (Table 3).

Length Group	N	Accuracy	Precision	Recall	F1
Short	699	0.61	0.62	0.55	0.58
Medium	677	0.64	0.60	0.51	0.55
Long	722	0.73	0.66	0.60	0.63

Table 3: *Qwen2.5-7B-Instruct* performance across review length groups.

5.4 Preliminary Results and Figures

The experimental results reveal two key patterns in fake review detection performance. First, the supervised *RoBERTa* baseline consistently outperforms the zero-shot *Qwen2.5-7B-Instruct* model across all evaluation metrics and review-length groups. This suggests that task-specific fine-tuning allows *RoBERTa* to better capture linguistic patterns associated with deceptive reviews, whereas the zero-shot Qwen model relies only on pre-trained language knowledge. Second, review length affects model performance. For both models, performance generally improves as review length increases, with the highest Qwen accuracy observed for long reviews (0.73 accuracy, 0.63 F1-score). Longer reviews provide richer contextual information that helps models identify deception cues. Short reviews contain limited context and are therefore more difficult to classify. For example, the short review “Great product, works perfectly.” was classified as genuine by the zero-shot LLM despite being labeled deceptive. Figure 2 compares *RoBERTa* and Qwen across review-length groups and shows that fine-tuned transformers remain more reliable for deception detection.

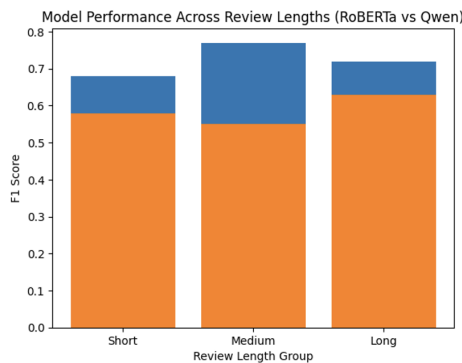


Figure 2: Model performance (F1-score) across review-length groups for *RoBERTa* and Qwen.

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